Step 1: Planning (Library Management System)

1. Idea

Idea: Develop a **Library Management System (LMS)** to replace the manual book borrowing process.

2. Problem Identification

Problem:

- Book borrowing is recorded manually.
- Students wait in long queues.
- Books are often misplaced.
- Librarians cannot quickly check book availability.
- Preparing reports takes a long time.

Goal:

Develop software that digitizes library operations and improves efficiency.

3. Feasibility Study

Evaluate whether the project is practical.

Technical Feasibility

- Can the university develop the software?
- Do developers have experience with databases and web development?
- Is the required technology available?
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Economic Feasibility

- Does the university have enough budget?
- Can it afford computers, servers, and software development costs?

Operational Feasibility

- Will librarians and students use the new system?
- Can staff be trained to operate it?

Schedule Feasibility

- Can the software be completed before the next academic semester?

Legal Feasibility

- Can student data be stored securely?
- Does the system comply with university privacy policies?

4. Cost Estimation

Estimate the project's expenses.
Example:

Item	**Estimated Cost (BDT)**
Software Development	8,00,000
Database Server & Hardware	2,00,000
Staff Training	1,00,000
Testing & Deployment	2,00,000
Total	13,00,000

5. Time Estimation
Estimate how long each activity will take.

Activity	**Duration**
Planning	1 week
Requirement Analysis	2 weeks
Design	2 weeks
Development	6 weeks
Testing	2 weeks
Deployment	1 week
Total	**14 weeks**

6. Team Formation
Assign roles to team members.

Role	**Responsibility**
Project Manager	Manage the project
Business Analyst	Gather requirements
UI/UX Designer	Design interfaces
Software Developer	Implement the system
Database Administrator	Design and manage the database
Tester (QA Engineer)	Test the software
System Administrator	Deploy and maintain the system

7. Project Plan (Final Output)

After completing all previous steps, prepare the project plan.

Example Project Plan

Project Name: Library Management System

Objective:

Digitize book borrowing and returning, reduce waiting time, and improve library management.

Scope:

- Student login
- Book search
- Borrow and return books
- Fine calculation
- Report generation

Estimated Budget: 13,00,000

Estimated Duration: 14 weeks

Project Team:

- 1 Project Manager
- 1 Business Analyst
- 2 Developers
- 1 Database Administrator
- 1 QA Engineer

Expected Outcome:

A web-based Library Management System that enables students and librarians to manage library resources efficiently.